Prabhavit Gupta

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Career Summary

- Automation Software Engineer holding dual degrees in Mechanical Engineering (B.E.) and Management Engineering (M.Sc., Politecnico di Milano).
- Solid foundation in programming, optimizing, and debugging PLC software for automated machinery across Siemens (TIA Portal) and Rockwell Automation platforms.
- Hands-on expertise interfacing field hardware including brushless motors, servo-drives, optical/inductive sensors, and pneumatic actuators with control logic.
- Experienced in electrical schematic interpretation, hardware-in-the-loop diagnostic testing, factory collaudo, and site commissioning.

Technical Skills

- **PLC Programming Platforms** – Siemens TIA Portal (S7-1200, S7-1500), Rockwell Automation / Allen-Bradley (Studio 5000 / RSLogix), Ladder (LAD), Structured Text (SCL)
- **Motion Control & Field Devices** – Servo Drives, Frequency Inverters, Stepper Motors, Encoders, Pneumatic Actuators, Industrial Sensors (Photoelectric, Proximity)
- **Industrial Communication** – PROFINET, PROFIBUS, Ethernet/IP, Modbus TCP/RTU, OPC UA, Fieldbus Couplers
- **Testing, Collaudo & Debugging** – FAT/SAT Protocols, Software Fault Isolation, Machine Commissioning, Root Cause Analysis, Electrical Schematics
- **Software & Tools** – Python, C++, Bash, SQL, Git, Linux, Technical Documentation

Visa & Language

- **Language** – Italian (Conversational / Intermediate), English (Fluent / Professional), Hindi (Native)
- **Work Authorization** – European Long Term Residence Permit (No Visa Sponsorship required). Driving License (Patente B). Available for on-site role in Milan.

Professional Experience

FlowSpace.co — *Automation Software & Systems Lead*
London, UK / Remote — *Mar 2024 – Ongoing*

- **Control Software Development & Optimization:**
  - Architected automated process control routines and continuous diagnostic suites for distributed machinery interfaces.
  - Developed high-speed telemetry handlers in Python and C++ monitoring drive response times, cycle parameters, and error states.
  - Authored technical machine specifications, functional descriptions, and debugging procedures for field engineers.
- **Hardware-Software Integration:**
  - Coordinated multi-protocol communication bridges interfacing field sensors and actuators with central supervisory logic.
  - Maintained 99.9% machine operational availability through proactive fault isolation and software logic refactoring.
- *Tools: Python, C++, Linux, Modbus, Industrial Networks, Docker, Git*

Urlink — *Systems Automation Engineer*
Milan, Italy — *Jul 2023 – Feb 2024*

- Conducted on-site machine testing, electrical schematic verification, and PLC program debugging during system collaudo.
- Configured servo-drive motion parameters, safety interlocks, and sensor inputs according to functional project specifications.

- Partnered with electrical engineering teams to resolve field wiring and communication anomalies during final commissioning.

UltraStrategy *Automation Developer*
Remote, Germany *Sep 2022 – Jun 2023*

- Developed automated scripts collecting machine operational data, cycle throughput, and diagnostic error codes.
- Structured equipment performance telemetry into comprehensive technical reports for quality assurance reviews.

Education

Politecnico di Milano *M.Sc. in Management Engineering (Digital Business & ICT)*
Milan, Italy *2020 – 2023*

Thapar Institute of Engineering and Technology
B.E. in Mechanical Engineering
Punjab, India *2014 – 2018*